



Association between sleep duration, demographic characteristics, health status, and lifestyle

A cross-sectional analysis based on NHANES 2017–2018 data

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CONTEXT



- Sleep is an essential biological process for physical and mental health.
- During sleep, the body and brain repair tissues, consolidate memory, and regulate hormones.
- Sleeping less or more than recommended has been associated with cardiovascular disease, diabetes, obesity, and cognitive impairment.
- Understanding these associations may support prevention strategies and healthier sleep habits.



**Sleep matters
across the lifespan**

HYPOTHESIS



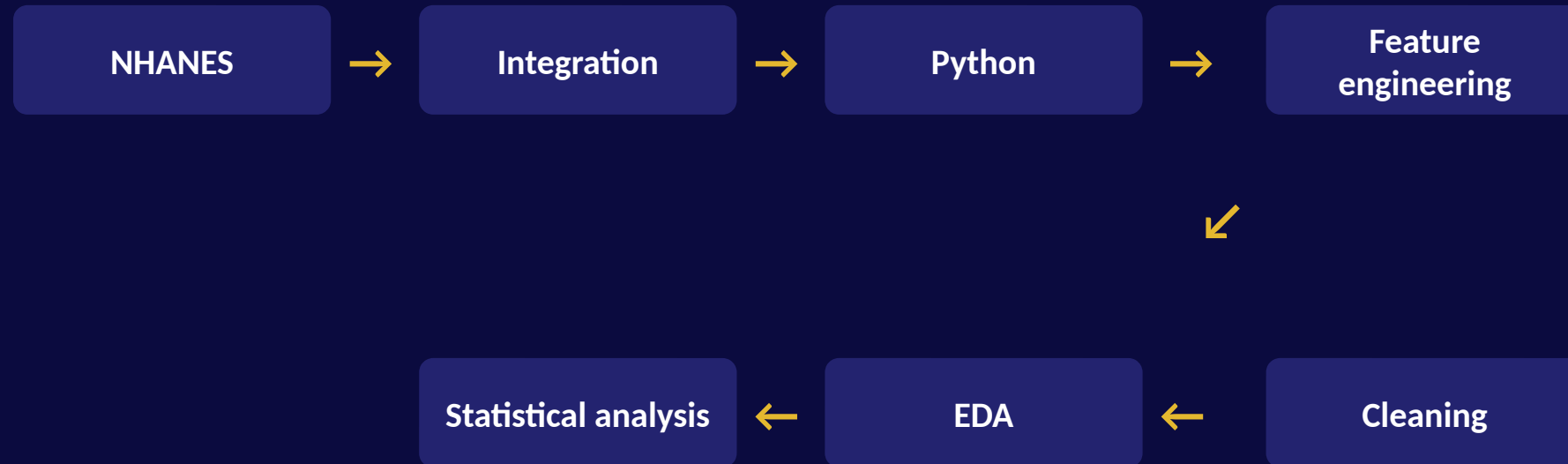
Sleep duration below or above the recommended range will be associated with a less favorable health and lifestyle profile.



METHODOLOGY



Project workflow



DATA PREPARATION



Data loading and renaming

Creation of categorical variables

Categorical variables were defined according to international clinical recommendations:

- WHO
- AASM
- ACC
- ADA

SAMPLE DESCRIPTION



4,103

Participants

51% / 49%

Women / Men

51.82

Mean age

2.53

Mean Poverty Index

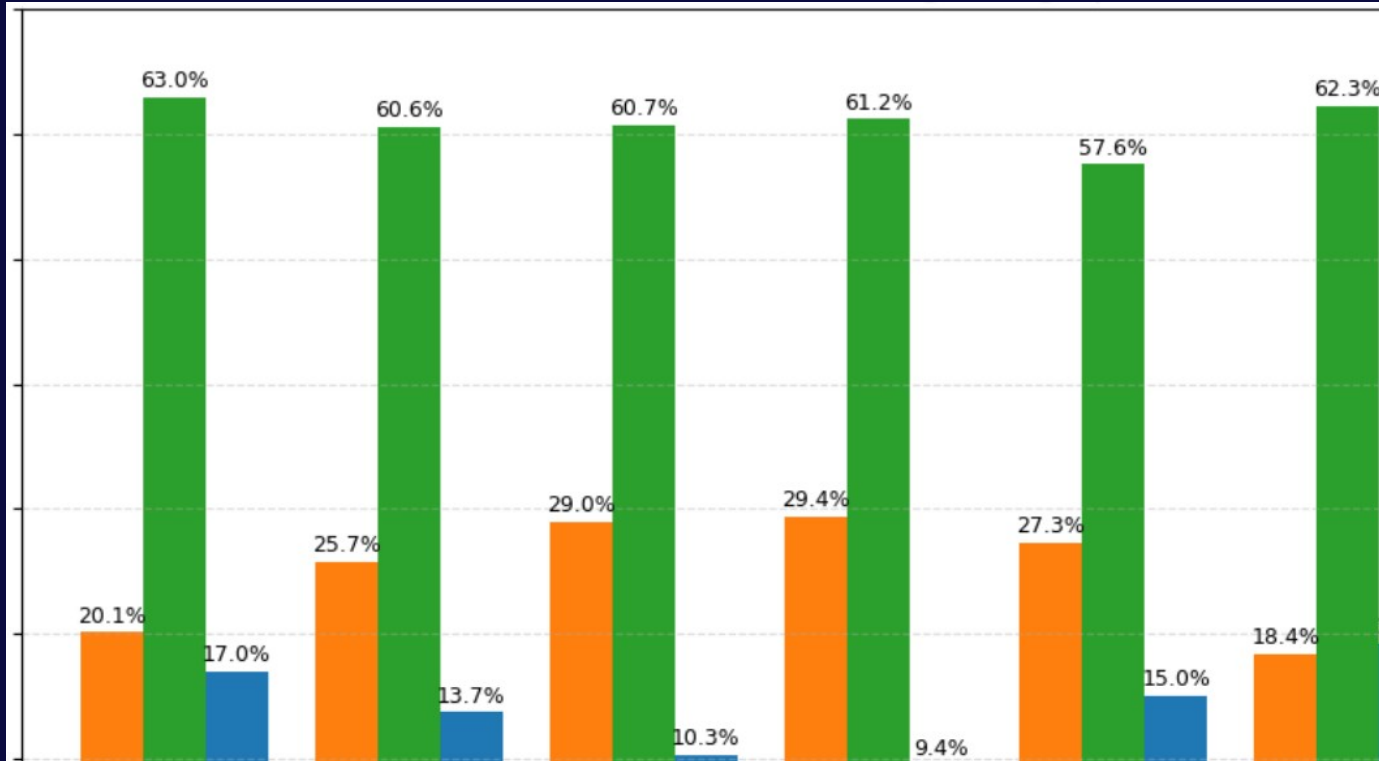
Good • Intermediate • Poor

Health profile

**Healthy • Intermediate •
Unhealthy**

Lifestyle profile

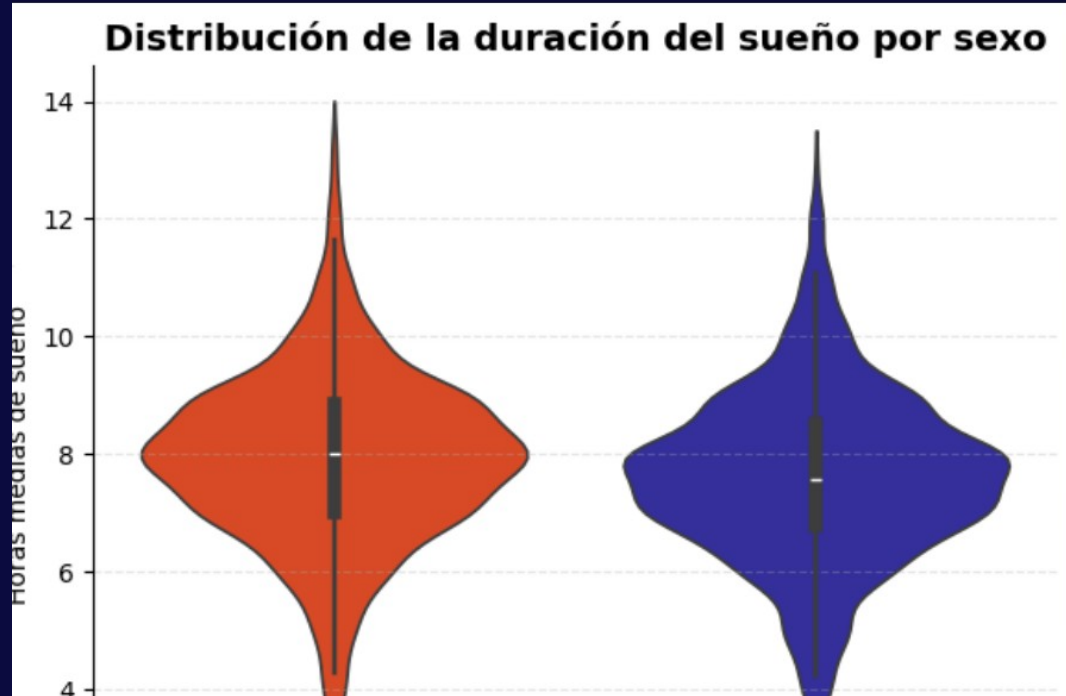
HOW DOES SLEEP VARY BY AGE?



- Recommended sleep was the predominant category in every age group (~60%).
- Short- and long-sleep categories showed opposite trends across age.

$\chi^2 = 40.51$ | $p < 0.001$ | Cramér's $V = 0.071$

HOW DOES SLEEP VARY BY GENDER?



Women reported a slightly longer mean sleep duration

18 minutes more sleep per night for women

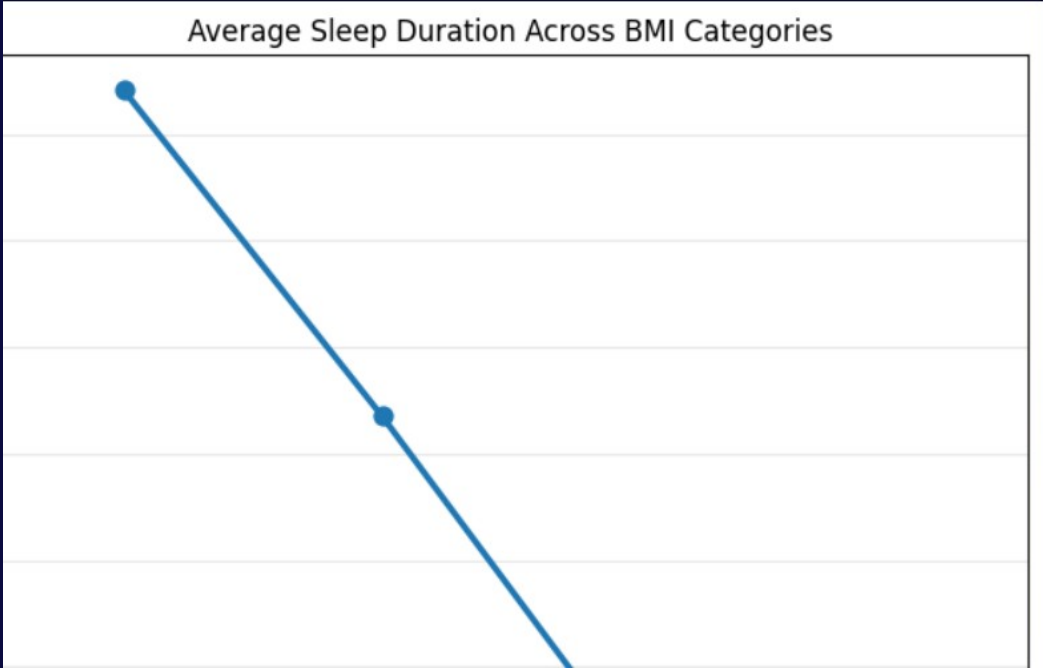
The overall sleep-duration distributions were very similar between sexes.



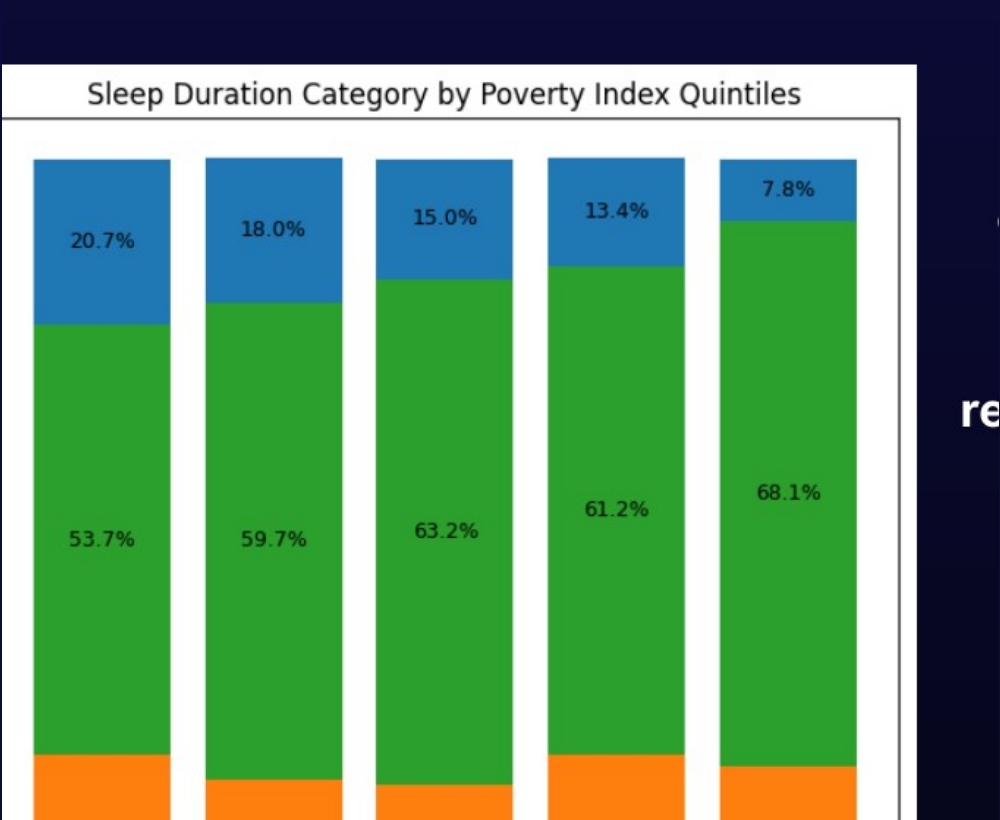
DO OTHER DEMOGRAPHIC FACTORS INFLUENCE SLEEP DURATION?



Does mean sleep duration differ by BMI?



Does sleep duration change by Poverty Index?



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BUILDING HEALTH AND LIFESTYLE PROFILES



HEALTH PROFILE

- BMI — body mass index
- HbA1c — glycemic control
- Blood pressure — systolic/diastolic
- HDL cholesterol
- Good health (1,964) · Intermediate (1,206) · Poor health (753)

LIFESTYLE PROFILE

- Physical activity — weekly exercise level
- Smoking — never / former / current
- Alcohol use — frequency and intensity
- Healthy (1,249) · Intermediate (1,893) · Unhealthy (961)

Scores were based on the percentage of the maximum possible score for the indicators available for each participant.

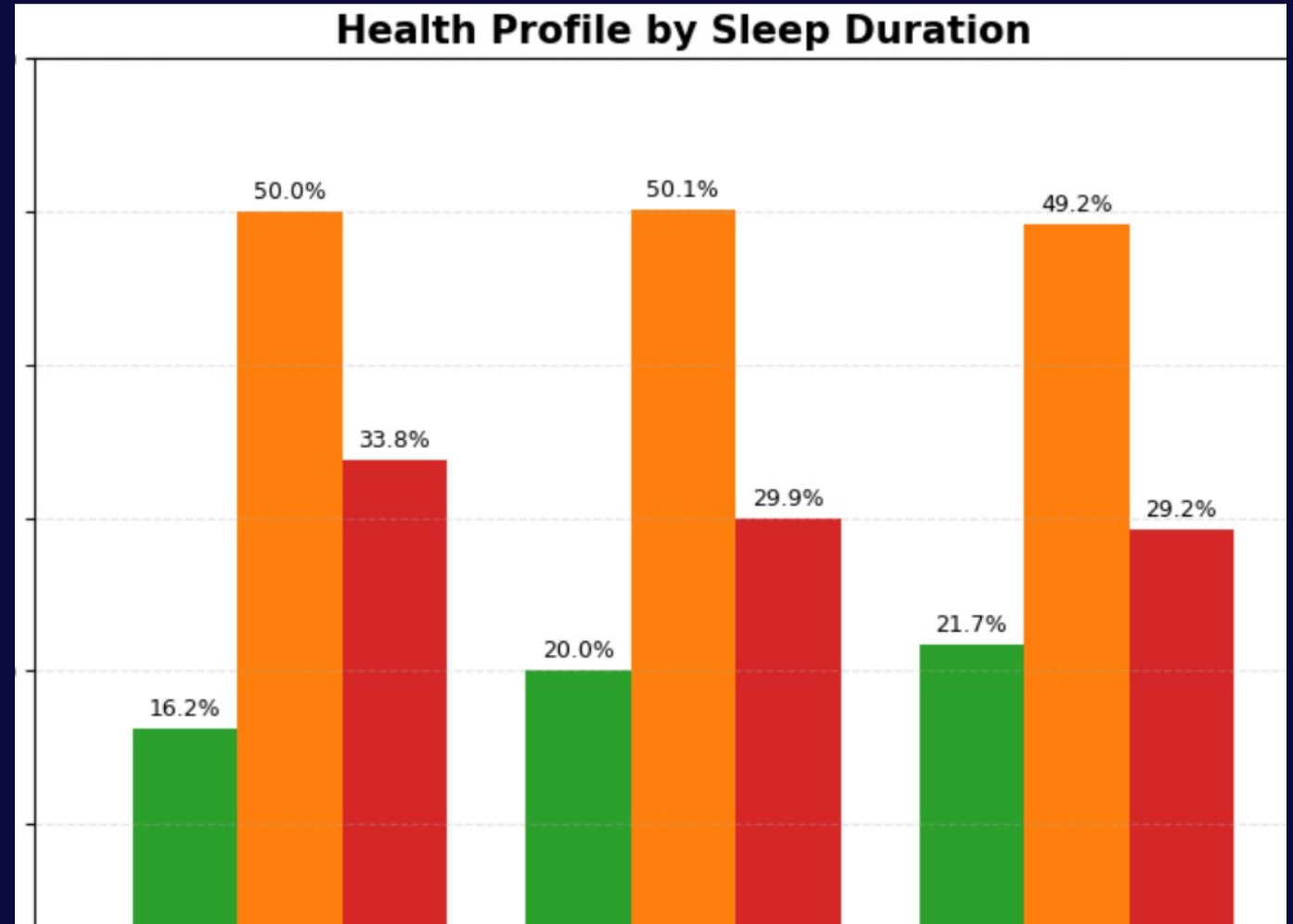
HOW IS SLEEP DURATION RELATED TO HEALTH PROFILE?



Short sleep included the highest proportion of participants with a poor health profile (33.8%).

Recommended and long sleep showed similar distributions.

$\chi^2 = 11.05$ | $p = 0.026$ | Cramér's $V = 0.038$



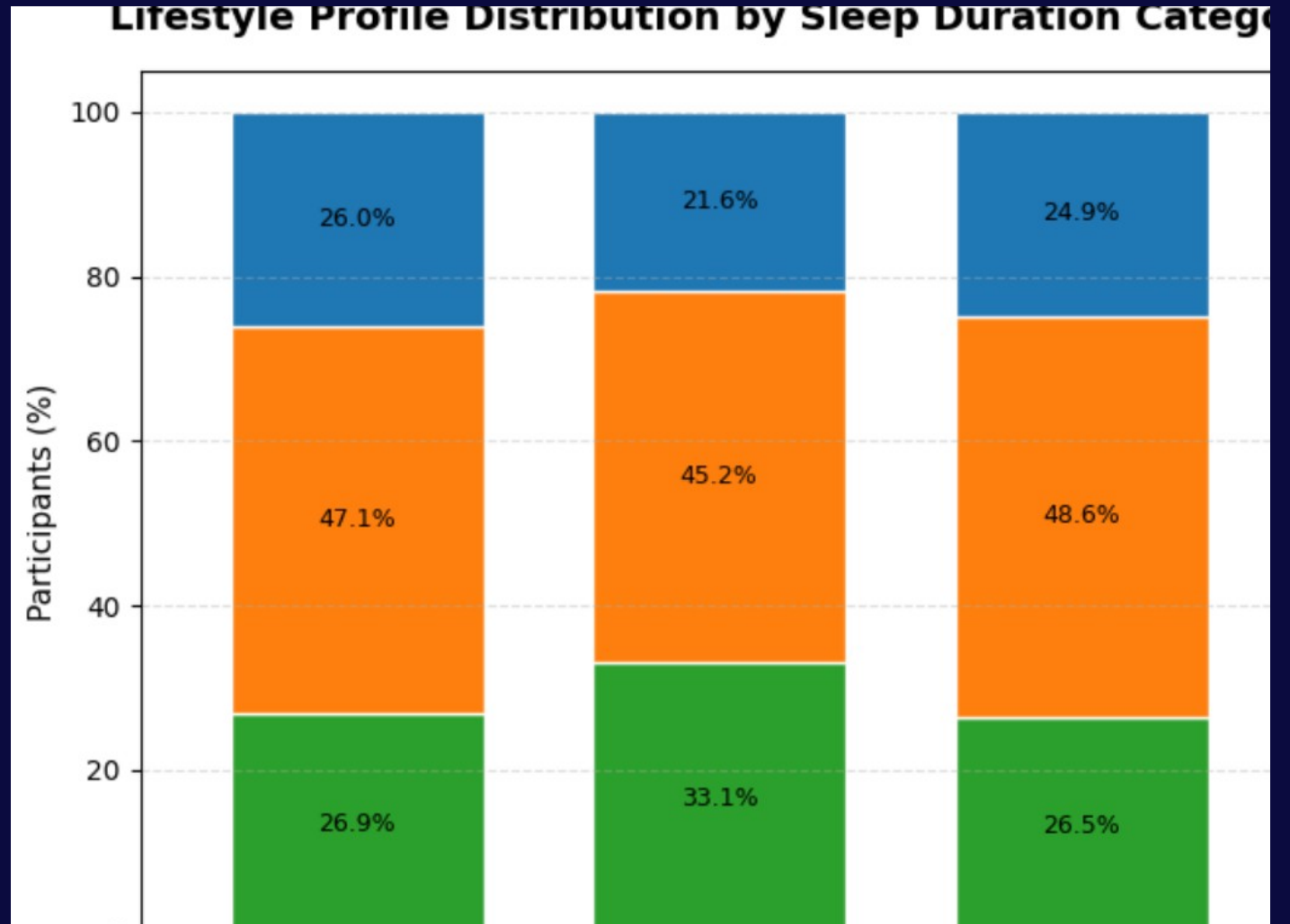
IS SLEEP DURATION RELATED TO LIFESTYLE?



Recommended sleep had the highest percentage of a healthy lifestyle profile (33.1%).

Short sleep had the largest proportion of participants with an unhealthy lifestyle (26.0%).

$\chi^2 = 20.92$ | $p < 0.001$ | Cramér's $V = 0.051$



SUMMARY OF STATISTICAL ASSOCIATIONS



All variables showed statistically significant associations with sleep duration ($p < 0.05$), but every effect size was very weak (Cramér's $V < 0.10$).

Variable	p-value	Cramér's V	Effect size
Age group	<0.001	0.071	Very weak
Poverty Index	<0.001	0.094	Very weak
Health profile	0.026	0.038	Very weak
Lifestyle profile	<0.001	0.051	Very weak

Poverty Index and age group showed the strongest associations, although their magnitude remained very weak.

CONCLUSIONS AND FUTURE WORK



CONCLUSIONS

- The hypothesis was partially supported.
- Sleep duration was significantly associated with demographic, health, and lifestyle factors.
- Low Cramér's V values indicate weak relationships and suggest that sleep duration is influenced by multiple factors.
- The findings reinforce the importance of healthy habits for better sleep quality and duration.

LIMITATIONS & FUTURE WORK

- The cross-sectional design does not establish causality.
- Diet, mental health, and some chronic diseases were not included.
- Future research could combine additional NHANES cycles, add variables, and apply predictive or longitudinal methods.



THANK YOU!

“If sleep does not serve an absolutely vital function, it is the greatest mistake evolution ever made.”

— Allan Rechtschaffen

Thank you for your attention
Questions?